D1: Forming a Group, Project Scope, and Product Perspective

Group Name: The Reservists

Group members:

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Project Name: Carrel

A campus study space booking system.

Problem the system is trying to solve:

University study space is a contested resource, this is especially true during midterms and final exam periods. Most places don't really have any sort of sign-up system, and the ones that do are usually just a patchwork of paper sign-up sheets outside individual rooms. The outcome is often wasted, or inefficient use of space.

Carrel is a single online booking system covering all bookable study spaces on campus. Students reserve rooms in advance, confirm their physical presence on arrival, and can cancel if plans change. Administrators define the room inventory, set the rules under which bookings are made, and resolve disputes through a dashboard. The system enforces conflict prevention so that two students cannot hold the same room at the same time, and it reclaims unused rooms automatically when the holder fails to check in within a grace window.

What the software product will do:

Use case 1: Reserve a room.

A student can browse available study spaces filtered by building, time, capacity, and other amenities, and they can reserve a slot within the rules the administrator has configured. Carrel will enforce things like per-student quotas (maximum hours per day, maximum concurrent bookings), prevent double-bookings, and other potential misuse. On confirmation, the student will receive a check-in code valid during a window around the slot start time, along with a calendar invitation containing the building, room number, and cancellation link.

Use case 2: Cancel a booking.

When a student arrives, they enter their check-in code or scan a QR code posted in the room, within the grace window (for example, the first ten minutes of the slot). A successful check-in marks the booking as held for its full duration. A booking with no check-in by the end of the grace window is released automatically and the room becomes available to others. A student may cancel a future booking at any time, freeing the slot immediately for other students. Cancellations made before the grace window do not count against any abuse threshold, but no-shows do, and accumulated strikes can be used by the administrator or by the system itself to throttle further bookings.

Use case 3: Administer rooms and booking rules.

A facilities administrator may manage a room's inventory, such as adding new bookable spaces, retiring rooms, marking maintenance closures, and editing amenity rules. Rules include but are not limited to per-student daily and weekly quotas, advance-booking horizon, minimum and maximum slot length, grace-window duration, no-show strike thresholds, and per-building blackout periods (e.g., during exam-period administration or facilities events). The administrator also reviews flagged bookings, resolves disputes through a dashboard, and may override the automatic enforcement (extend a slot, waive a strike, force-cancel a booking) when circumstances demand.

The benefits/objectives/goals of the software:

The primary goal is to make study-space allocation as easy and as fair as possible. Concretely, one thing Carrel aims to eliminate are ghost reservations through automatic check-in enforcement, so an unclaimed room returns to the pool almost immediately rather than sitting empty for its full booked block. It also gives students a single central place to find space, replacing older, more inefficient methods of administration such as patchwork of paper sheets or sending individual emails. Carrel will also surface usage data to facilities staff so they can identify under-utilized rooms, peak-demand periods, and mismatches between advertised amenities and what students actually book around. Because there is now an audit trail for disputes, a claim of "I had this room" can be resolved against the booking record rather than against word of mouth, or the last sign-up sheet to survive on the door.

Product Context and Origin

Carrel is an entirely new product, not a successor to or replacement for any specific existing university system. Comparable booking software exists at peer institutions — most commonly LibCal's room module at libraries running SpringShare, and in-house tools at some larger universities, but in practice these are scoped to a single building or department and do not unify allocation across an entire campus.